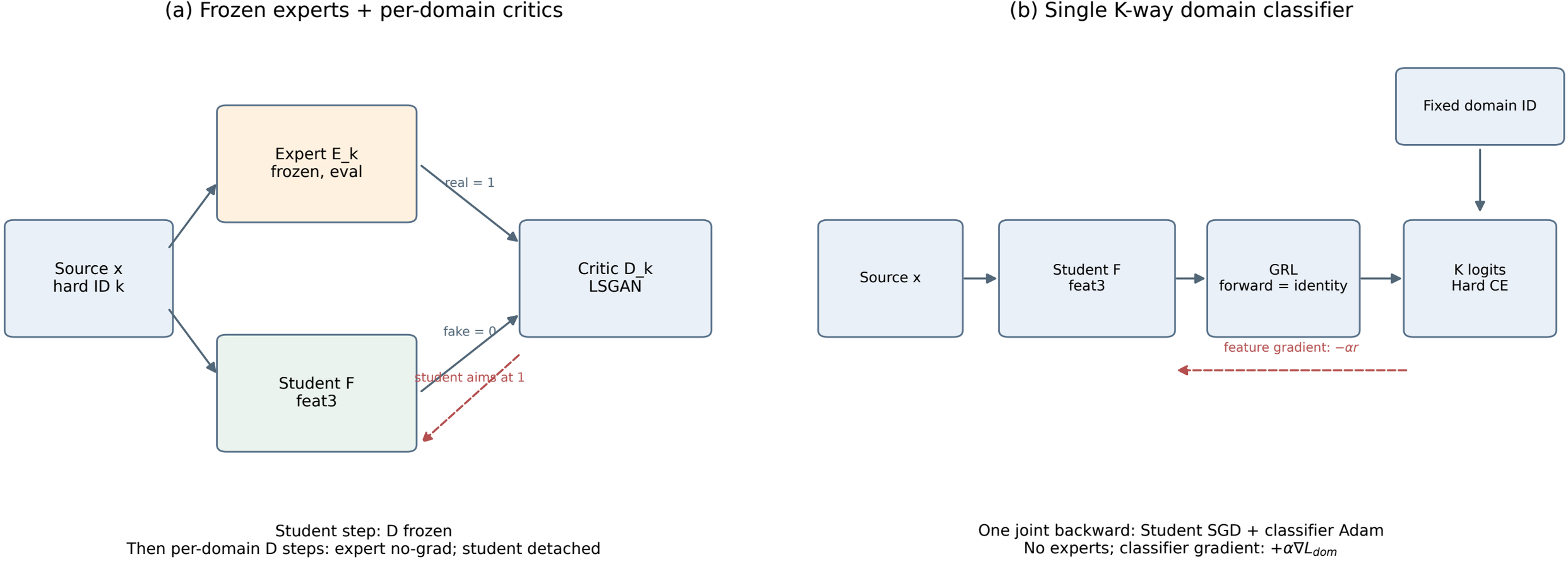
(a) Frozen experts + per-domain critics
Expert E_k
frozen, eval
Source x
hard ID k
Student F
feat3
Critic D_k
LSGAN
real = 1
fake = 0
student aims at 1
Student step: D frozen
Then per-domain D steps: expert no-grad; student detached
(b) Single K-way domain classifier
Fixed domain ID
Source x
Student F
feat3
GRL
forward = identity
K logits
Hard CE
feature gradient: −αr
One joint backward: Student SGD + classifier Adam
No experts; classifier gradient: +α∇L_dom
Shared task path: detection + 0.3 object Triplet (feat3 / feat2). Inference: detector only.